Ping-Jung Wu

Education

M.Sc. in Aerospace **10/2024 – present**
Technical University of Munich (TUM), Munich, Germany
Current GPA: 1.5
Key Focus: UAV Control, Trajectory Optimization, Autonomous Navigation
Selected Projects:

- **Drone SLAM Navigation**: Implemented visual mapping and navigation architecture for drone tunnel search
- **Rocket Altitude Control**: Designed simulations in a gas-dynamic system by implementing an MPC controller for altitude stabilization
- **Physics-Informed AI**: Implemented deep learning models for solving PDEs in solid mechanics, heat transfer, and traffic-flow dynamics

Related Courses: Investment and Financial Management, Aircraft Trajectory Optimization, Optimal Control and Decision Making

B.Sc. in Mechanical Engineering **09/2020 – 01/2024**
National Taiwan University (NTU), Taipei, Taiwan
GPA: 3.97/4.30
Key Focus: Linear Control, Fluid Mechanics
Related Courses: Flight Dynamics, Viscous Flow, Linear Control Systems

Research Experience

Master Thesis, Technical University of Munich **05/2026 –**

Coordinated Autonomous Exploration in Forests for Decentralized Multi-UAV Systems

- Develope a ROS2/C++ decentralized autonomous stack featuring ad-hoc meeting and spatial allocation protocols for multi-agent coordination in dense canopy search and mapping
- Demonstrate 15% reduction in mission time and a 14% decrease in average path length compared to baseline peer research across varying forest densities in simulation
- Preparing PX4 autopilot integration for real-world swarm flight validation

Semester Thesis, Technical University of Munich **10/2025 – 04/2026**

- Validated quadrotor's trajectory optimization methods from the frontline research
- Designed minimum control based optimizers under safe flight corridors
- Developed unified collision-avoidance constraints for a multi-agent system within a centralized architecture

Research Student, National Taiwan University **02/2022 – 07/2022**

- Constructed a control architecture by using setpoint tracking for tandem wing tail-sitter
- Implemented a cascaded PID controller for longitudinal control
- Validated aircraft dynamic responses through CFD simulations and longitudinal stability analysis

Personal Projects **07/2021 – 07/2023**

- **Attitude and Heading Reference System:** Developed an Arduino-based attitude and heading reference system using an Extended Kalman Filter for sensor fusion.

- **Fixed Wing Aircraft:** Constructed a monoplane by utilizing Thin Airfoil Theory
- **Tandem Wing Tail-sitter:** Built a concept aircraft with the capability for both vertical and horizontal takeoff and landing

Skills

Programming: Python, PyTorch, C / C++, ROS 2, MATLAB, Git
Languages: Mandarin (native), English (TOEFL 100 / 120), German (B2, currently improving), Japanese (basic)
Software: Ansys Fluent, Autodesk Inventor, AutoCAD, SolidWorks

Awards

Dean's List Award, National Taiwan University **Spring 2022, Spring 2023**

- Awarded top five percent of students for excellent academic performance

Masterpiece Award for Bachelor's Thesis, National Taiwan University **2023**

- Awarded by the College of Engineering for excellent research work

Club Activities

Software Developer, Horyzn, Technical University of Munich **05/2026 – present**

- Integrate and deploy UAV trajectory-planning algorithms within PX4 simulation
- Develop a vision-assisted waypoint navigation system with obstacle detection

Lecturer, NTU Aviation Club, National Taiwan University **09/2022 – 12/2023**

- Subjects: Basic Aerodynamics, Model Aircraft Manufacturing